SIMBLOCKS // RESULT

2026-10-16 02:18Z

RIGID BLADE TIP LOAD

PASS VERIFIED

What is the most aggressive tip load to apply to a rigid Flapjack blade, from the flexible-blade sims?

WORST BENDING IMPULSE AT TIP	INFO
7.85	N s

peak 28.7 kN in-plane; worst out-of-plane 2.81 N s, 5884 N; current load 2.30 N s

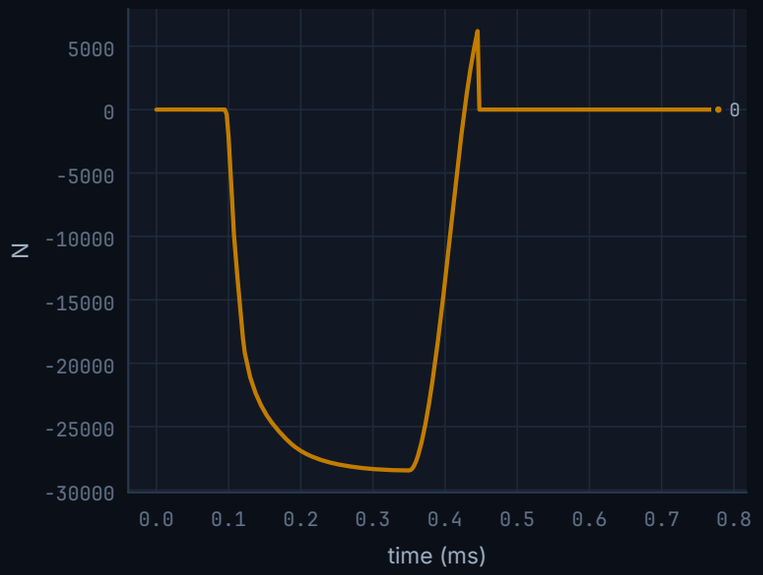
No failure modes defined: ask the user what failure means for this part.

Rigid blade tip load: VERIFIED | PASS 12 WARN 1 FAIL 0 SKIP 0 | 37 runs, 13.6 s

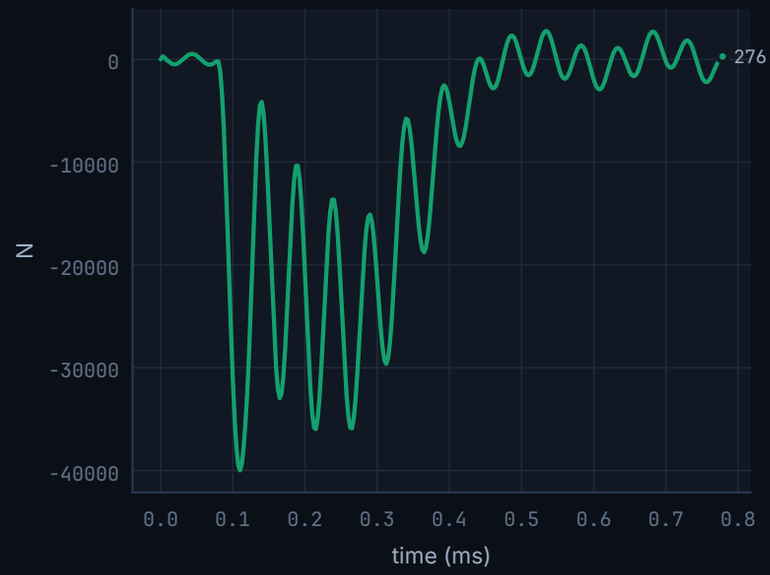
RIGID BLADE TIP LOAD

Time histories of the governing states (nominal case)

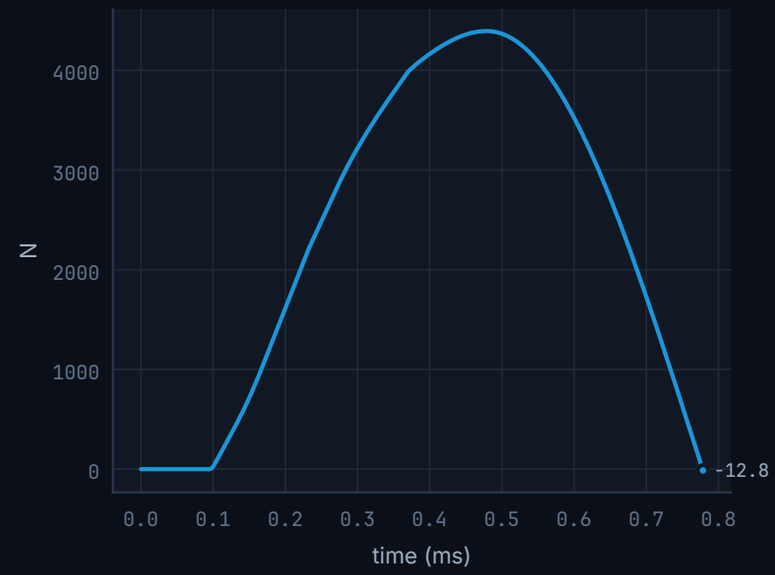
Worst overall: Y in-plane



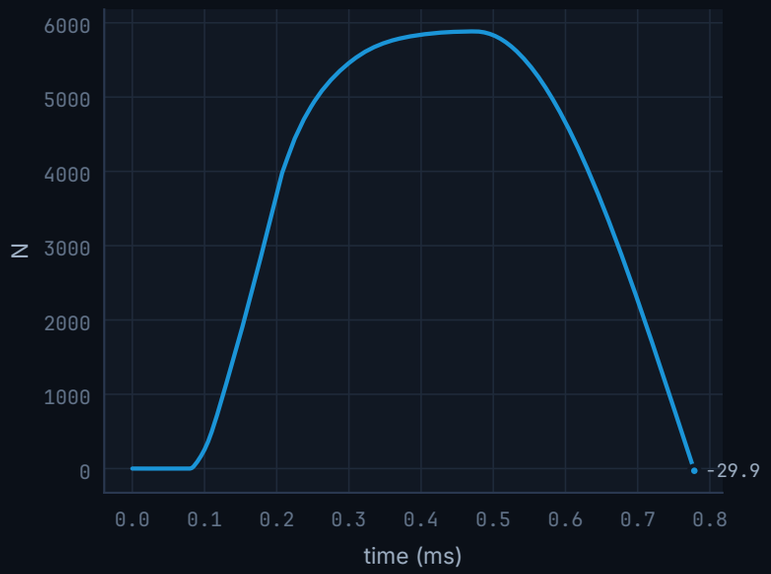
Worst overall: X radial (centrifugal removed)



Worst overall: Z out-of-plane



Worst out-of-plane case: Z



RIGID BLADE TIP LOAD

What to design around: drivers of the governing margin and the design-variable sweep

SIMBLOCKS // VERIFICATION

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RIGID BLADE TIP LOAD

Automated verification battery

● PASS VERIFIED

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	1 bounds respected
stability_margin	● PASS	critical dt 4.2e-06 s / dt 2.5e-07 s = 16.8x
dt_convergence	● WARN	tip_J_lat: err 0.029% (order 4.1); tip_F_lat_pk: err 0.013% (order 0.47); tip_Fz_pk: err 0.28% (order 2.4); tip_J_x: err 0.4% (order 3.5)
energy_ledger	● PASS	worst closure residual 0.47% across 3 dt levels (tol 5%)
limit:robots never meet (1 ms run) → no tip load	● PASS	matches expected limit
knockout:rigid H blade	● PASS	tip_J_lat +2.67%, tip_F_lat_pk +626%, tip_Fz_pk +1.59e+03%, tip_J_x +5.43%
knockout:no blade plasticity	● PASS	tip_J_lat +61.8%, tip_F_lat_pk +186%, tip_Fz_pk -23.1%, tip_J_x -58.5%
knockout:no blade friction	● PASS	tip_J_lat -72.9%, tip_F_lat_pk -11%, tip_Fz_pk -79.4%, tip_J_x +68.8%
knockout:no gyroscopic coupling	● PASS	tip_J_lat +0.00374%, tip_F_lat_pk +0.000733%, tip_Fz_pk +0.0161%, tip_J_x +0.00775%
sensitivity	● PASS	9 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	10 samples, 0 failed finite/bounds/ledger (0.0%)